



DPUSampleFilePlayer

1. Inputs:

<i>Start</i>	Input to start the sample
<i>In[1...50]</i>	Inputs for the individual channels

2. Dynamic parameters:

<i>Autorepeat</i>	Flag, which indicates if the sample – started once – should be permanently repeated (<i>not possible</i> in combination with <i>ReadOnDemand</i>)
<i>SampleRateMS</i>	Actual sample rate of the original sample
<i>Multiplicator[1...50]</i>	Multiplication factor for conversion of the channels. Default: 1
<i>Divisor[1...50]</i>	Attenuation factor for the channels. Default: 1
<i>Offset[1...50]</i>	Offset alignment for the channels. Default: 0
<i>Default[1...50]</i>	Default value of the channels for time steps outside of the sample. Default: 0

3. Static parameters:

<i>ReadOnDemand</i>	Read of file only on demand during simulation
<i>SampleID[1...50]</i>	Column assignment of the channel
<i>FileName</i>	File name of data sample. File name can be an absolute path (e.g. "C:\MySamples\Samples.txt") or it can be relative to the SILAB install directory (e.g. "Data\Sample1.asc").
<i>FileRowStart</i>	Startline in data file for data
<i>FileRowEnd</i>	Endline in data file for data
<i>Dezimal</i>	Decimal separator
<i>Default value '.'</i>	
<i>CR</i>	Separator for line break.

4. Outputs:

<i>SamplePlayTime</i>	Elapsed playing time of sample
<i>Out[1...50]</i>	Output channels

5. Launchdata:

<i>SFP_FILENAME</i>	LaunchData token to source the FileName from the launch data dialogue. Overwrites the original FileName value.
---------------------	--

6. Functional principle:

DPUSampleFilePlayer can read a data file in text file format. This file can be replayed as often as you want. The inputs are calculated with the sample data as following:



$$Out_x = \frac{(In_x + Sample_x(t)) \cdot Multiplier_x}{Divisor_x} + Offset_x$$

$Sample_x(t)$ represents the value of the file, which is valid at time t . With every simulation step the next line of the data file is output.

The data file can be played slower, as well. Parameter *SampleRateMS* represents the original sampling rate in milliseconds with which the sample has been recorded. If the original sampling rate is larger than the time interval of the DPU, the values are repeated, until in mean the original sampling rate is reached. However, if the sample rate is less than the timing of the DPU, then the sample is still played with the timing of the DPU.

The DPU reads the file in the preparation step. It reads the file line by line, starting from the specified startline (*FileRowStart*) until endline (*FileRowEnd*) and analyzes the rows to find the columns. As separator every character is counted, which does not occur in valid numbers. Separators of the first row are ignored. Any number of separators is interpreted as separators to the next line between the columns. The column found are associated with the outputs.

Not all columns have to have the same length. However, it could happen, that incorrect assignments of values are made, in case of different length of columns.

Example for a table which should be read

```
--- Begin of Datafile ---
Time(Def_Tact)    SteeringWheelAngle[Grad]    YawRate[deg/s] ...
91.63 -1.189648437    0.144140625 ...
91.64 -1.8    0.241796875 ...
91.65 -1.8    -0.00234375 ...
91.66 -1.8    -0.148828125    ...
91.67 -1.8    -0.00234375 ...
91.68    -2.410351562    -0.051171875    ...
91.69    -1.8    -0.1    ...
91.7 -1.189648437    0.046484375 ...
91.71    -1.8    -0.148828125    ...
91.72    -2.410351562    -0.344140625    ...
91.73    -2.410351562    -0.344140625    ...
91.74    -1.8    -0.246484375    ...
...
--- End of Data ---
```

The actual numeric values start in line 3. The value of *FileStartRow* will be set to 3. As output *Out1* the yaw rate in rad/s and as output *Out2* the steering wheel angle in rad with reversed algebraic sign should be output. In addition, as output *Out3* the steering wheel angle in degrees, albeit with same algebraic sign as output *Out2*, shall be output.



SILAB DRIVING SIMULATION

version Version 7.1 documentation

DPUFileSamplePlayer Sample

```
{  
    FileRowStart = 3;  
    # Channel 1 : yaw rate of column 3 (of data file)  
    SampleID1 = 3;  
    Multiplicator1 = 3.1415;  
    Divisor1 = 180.0;  
  
    # Channel 2 : steering wheel angle of column 2 (of data file)  
    SampleID2 = 2;  
    Multiplicator2 = -3.1415;  
    Divisor3 = 180.0;  
  
    # Channel 3 : steering wheel angle of column 3 (of data file) in  
    # degrees  
    SampleID3 = 2;  
    Multiplicator3 = -1.0;  
};
```